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B.Tech. Degree IV Semester Examination April 2018

ME 15-1406 MANUFACTURING PROCESSES (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

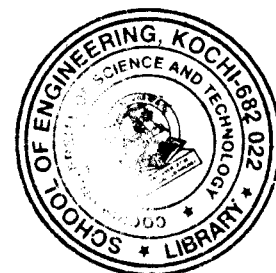
(10 × 2 = 20)

- I. (a) 'Risers in sand casting should solidify at a slower rate than casting cavity'. Justify the statement.
- (b) Compare the advantages and disadvantages of sand casting with die casting process.
- (c) What are the characteristics of an ideal molding sand that is to be used in sand casting process?
- (d) Explain the process of hot chamber die casting.
- (e) Explain the process of shell molding with a suitable figure.
- (f) Distinguish between punching and blanking processes in sheet metal operations.
- (g) State the advantages and disadvantages of direct and indirect extrusion.
- (h) How can spring back be avoided in bending operations?
- (i) Explain the process of thermit welding.
- (j) Differentiate between brazing and soldering.

PART B

(4 × 10 = 40)

- II. (a) Explain the different methods to prevent impurities and turbulence in sand moulds. (5)
 - (b) Explain the different types of patterns used in sand casting. (5)
- OR**
- III. (a) With a neat figure, explain the nomenclature of a sand casting mould, showing all the elements. (6)
 - (b) What do you mean by directional solidification? What are the methods for achieving directional solidification? (4)
- IV. (a) What is investment casting? Explain with necessary figures. (5)
 - (b) What are the product design considerations to be followed in order to reduce defects arising in a sand cast product? (5)

OR**(P.T.O.)**

- V. (a) Explain centrifugal casting and semi centrifugal casting processes bringing out their differences. (7)
- (b) What are the methods for reducing the cost of a sand cast product? (3)
- VI. (a) Differentiate between coining and embossing processes in sheet metal operations. (4)
- (b) What do you understand by hot working and cold working? Give their advantages and disadvantages. (6)
- OR**
- VII. (a) Explain different types of forging operations. (5)
- (b) How is swaging process performed? (3)
- (c) After severe cold working operation, the material/product needs heat treatment process. Justify the statement. (2)
- VIII. (a) Explain different types of resistance welding processes with necessary figures. (8)
- (b) Compare the advantages and disadvantages of AC current and DC current arc welding. (2)
- OR**
- IX. Explain the following welding processes.
- (a) Gas metal arc welding. (3)
- (b) Electron beam welding. (4)
- (c) Friction welding. (3)
